PROJECT REPORT

A Comparative Study of Spatial, Frequency, and Hybrid Feature Learning for AI Generated Video Detection

Submitted by

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CERTIFICATE

This is to certify that the Mini Project work titled "A Comparative Study of Spatial, Frequency, and Hybrid Feature Learning for AI Generated Video Detection" submitted by G. Karthik Yadav (100523729022), K. Pujyank (100523729031), and M. Sriharshith (100523729037), students of the Department of Computer Science and Engineering, University College of Engineering, Osmania University, is a record of the bonafide work carried out by them during the academic year 2025–26.

The project is submitted in fulfillment of the requirements for the Bachelor of Engineering in Computer Science and Engineering award.

The work has been done under my supervision and, to the best of my knowledge, is original and reflects the students' sincere efforts.

Signature of the Supervisor

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Signature of the Head of the Dept

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DECLARATION

We, G. Karthik Yadav (100523729022), K. Pujyank(100523729031), and M. Sriharshith (100523729037), students of the Department of Computer Science and Engineering, University College of Engineering, Osmania University, hereby declare that the work presented in this Mini Project titled "A Comparative Study of Spatial, Frequency, and Hybrid Feature Learning for AI Generated Video Detection" is an original contribution carried out by us during the academic year 2025–26.

This project report is submitted in fulfillment of the requirements for the degree of Bachelor of Engineering in Computer Science and Engineering specializing in Artificial Intelligence and Machine Learning. It has not been submitted elsewhere to award any degree or diploma.

We affirm that no part of this report is plagiarized, and that wherever references have been made, they have been appropriately cited. The findings and analysis presented are based on our genuine and authentic work carried out under the guidance of Prof. Dr. V. B. Narsimha.

We further declare that we have adhered to ethical practices throughout the research and project development process, maintaining academic integrity at every stage. The data, analysis, and outcomes of this report are factual to the best of our knowledge, and we take full responsibility for the contents of this submission.

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1. Abstract

With the rapid growth of AI generated media, distinguishing between real and synthetic videos has become an important challenge in digital forensics and media authentication. This project presents a deep learning based approach for detecting AI generated and real videos using spatial and frequency domain analysis.

Unlike traditional methods that rely only on RGB images, this work uses the Fast Fourier Transform (FFT) to extract frequency domain features that capture subtle artifacts often present in AI generated content. A ResNet18 based convolutional neural network is used to classify video frames through three comparative approaches: RGB based analysis, FFT based analysis, and a fusion of RGB and FFT features.

The fusion approach combines spatial and spectral information to improve robustness and detection accuracy, particularly across varying video qualities and compression levels. Experimental results show that incorporating frequency domain analysis improves performance compared to using only spatial features.

The proposed method demonstrates strong potential for applications such as digital forensics, content moderation, and media authentication, where reliable detection of synthetic content is essential.

2. Introduction

The rapid advancement of generative artificial intelligence, especially through deepfake generation models and Generative Adversarial Networks (GANs), has made distinguishing between real and synthetic media increasingly challenging. Modern AI models can generate highly realistic videos that closely imitate human faces, expressions, and environments, making manual verification unreliable in many cases.

Traditional deepfake detection methods mainly focus on spatial features such as facial inconsistencies, texture patterns, and visible artifacts in images or video frames. While these approaches have shown promising results, their effectiveness decreases as generative models become more advanced and capable of producing visually convincing outputs with fewer detectable distortions.

Recent studies indicate that AI generated content often carries hidden artifacts in the frequency domain due to convolutional operations, upsampling methods, and generation architectures used in synthetic media creation. Although these artifacts may not be easily visible in the spatial domain, they can be identified using signal processing techniques such as the Fast Fourier Transform (FFT).

Motivated by these observations, this project explores the use of frequency domain analysis along with deep learning for detecting AI generated videos. A comparative study is performed using three approaches: RGB based analysis, FFT based analysis, and a fusion of both RGB and FFT features. By combining spatial and spectral information, the proposed system aims to improve detection accuracy, robustness, and the reliable separation of real and synthetic video content.

3. Problem Statement

The rapid development of artificial intelligence has enabled the creation of highly realistic synthetic videos, making it increasingly difficult to distinguish AI generated content from authentic videos. Technologies such as deepfakes and generative models can replicate facial expressions, movements, and visual details with remarkable accuracy, creating serious concerns related to misinformation, identity misuse, digital fraud, and media manipulation.

Traditional detection methods mainly focus on spatial features such as texture inconsistencies, facial irregularities, or visible pixel level artifacts. Although these approaches have shown good performance in earlier deepfake detection tasks, they often struggle when dealing with advanced generative models that produce visually coherent outputs with very few noticeable distortions. As AI generated videos continue to improve, relying solely on spatial information is becoming insufficient for reliable detection.

Another major challenge is that many synthetic videos contain hidden artifacts introduced during generation, especially due to convolutional operations, upsampling methods, and model architectures used in generative networks. These artifacts may not be easily observable in normal RGB images but can often be identified in the frequency domain. However, many existing detection systems do not effectively utilize this information.

Therefore, the problem addressed in this project is how to build a robust and accurate system for distinguishing real and AI generated videos by leveraging both spatial and frequency domain information. This project investigates whether frequency based analysis using Fast Fourier Transform (FFT), either independently or combined with RGB features, can improve detection performance compared to traditional spatial only approaches.

To address this, a comparative study is carried out using three methods: RGB based detection, FFT based detection, and a fusion of RGB and FFT features to analyze their effectiveness and determine whether combining both representations provides a more reliable solution for AI generated video detection.

4. Literature Review

Research in AI generated media detection has evolved significantly, with various approaches proposed to distinguish synthetic content from real media. Existing studies can broadly be viewed through the perspective of the three approaches considered in this project: **RGB based methods, FFT based frequency domain methods, and fusion based approaches combining both.**

Early deepfake detection research mainly focused on **RGB based or spatial domain analysis**, where convolutional neural networks(CNNs) were trained to identify visual artifacts such as facial inconsistencies, unnatural textures, blending errors, and pixel level distortions. Models based on deep learning architectures showed promising performance on benchmark datasets by learning discriminative spatial patterns from individual frames. These methods formed the foundation for many modern detection systems and demonstrated that spatial features could effectively detect manipulated content.

However, as generative models such as GANs became more advanced, the limitations of purely RGB based methods became apparent. High quality synthetic videos often contain very few visible distortions, making them difficult to detect using spatial features alone. This highlighted the need for alternative approaches capable of identifying artifacts beyond what is visible in the image domain.

To address this challenge, researchers began exploring **frequency domain analysis**, leading to growing interest in **FFT based detection methods**. Studies have shown that AI generated media often carries hidden spectral artifacts introduced by convolutional layers, up sampling processes, and generator architectures. By transforming images into the frequency domain using Fast Fourier Transform (FFT), these patterns can be analyzed and used for classification. Several recent works have reported that frequency based models can detect synthetic traces that may remain unnoticed in RGB images, making this approach an important advancement in deepfake detection research.

Alongside spatial and frequency methods, some research has investigated combining multiple feature domains. These **fusion based approaches**, which integrate RGB and FFT information has gained attention because they utilize the strengths of both representations. While RGB analysis captures visual and structural information, FFT analysis captures hidden spectral patterns, and together they provide a more complete understanding of the content. Recent studies suggest that such hybrid models often outperform single domain models in robustness and detection accuracy.

Although temporal methods such as RNNs, LSTMs, and 3D CNNs have also been explored to analyze motion based inconsistencies, they often require higher computational cost and complex sequence modeling. In contrast, the comparative framework adopted in this project focuses specifically on analyzing and evaluating **three key detection strategies: RGB, FFT, and Fusion** to understand their individual strengths and determine whether combining spatial and frequency domain features leads to improved detection of AI generated videos. This forms the foundation and motivation for the proposed work.

5. Research Gap

Although significant progress has been made in detecting AI generated media, several research gaps still exist, particularly when comparing and combining different detection strategies such as **RGB based analysis, FFT based analysis, and fusion based approaches**, which form the focus of this project.

Most existing deepfake detection methods rely heavily on **RGB based spatial features**, where models learn visible artifacts such as texture inconsistencies or facial irregularities. While these methods have shown strong results, they often struggle as synthetic media becomes more realistic. This creates a gap in relying solely on spatial domain information for robust detection.

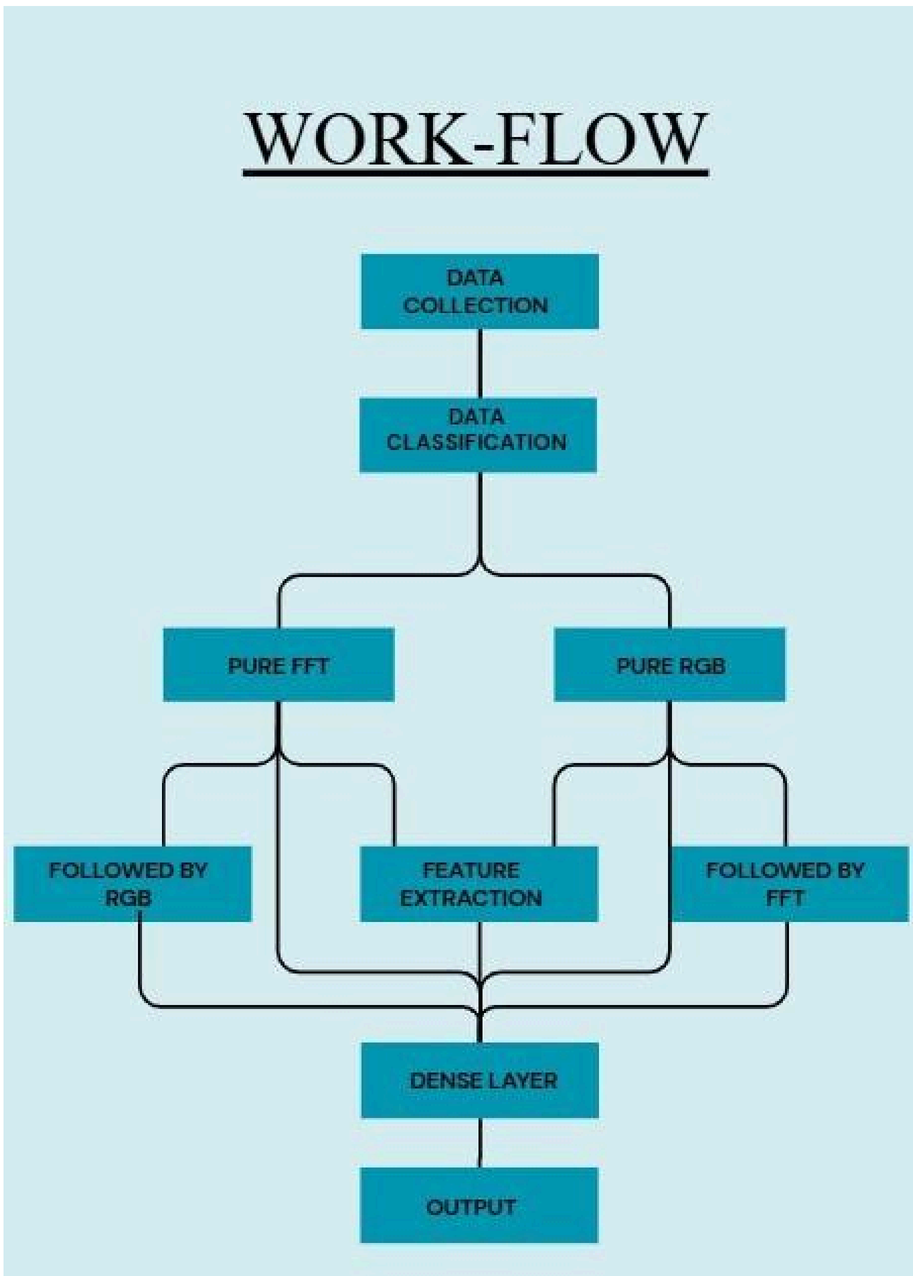
At the same time, **FFT based frequency domain methods** have shown that synthetic content often carries hidden spectral artifacts introduced during generation. However, much of the existing work in this area focuses mainly on image level detection, and comparatively less research has explored how frequency domain features can be effectively used for **video based classification** using modern deep learning models. This reveals a gap in leveraging frequency information for practical video detection systems.

Another important gap is limited exploration of **fusion based methods**. While some studies suggest that combining spatial and frequency domain information can improve detection, end to end frameworks that systematically compare **RGB, FFT, and their fusion** remain underexplored. There is still a need to better understand whether combining these two feature domains consistently outperforms individual approaches and how such fusion can be implemented efficiently.

Moreover, many existing models are computationally complex and difficult to deploy in real world or resource constrained environments. There is a need for approaches that maintain a balance between detection performance and practical efficiency.

This project attempts to address these gaps through a comparative study built around the three selected methods. It explores **RGB based detection** as a baseline spatial approach, **FFT based detection** to capture hidden frequency artifacts, and a **fusion approach** that combines both representations for improved feature learning. By using a lightweight yet effective pretrained **ResNet18** architecture and a practical training pipeline, this work aims to contribute toward a more efficient and robust framework for AI generated video detection.

6. Methodology and Workflow



The methodology adopted in this project follows a structured pipeline for detecting whether a video is **AI generated or real**, while performing a comparative study across the **three selected methods: RGB based detection, FFT based detection, and Fusion (RGB+FFT) detection (version 1 2)**. The workflow is designed to evaluate each method individually and analyze how combining spatial and frequency domain information impacts performance.

- **Plain RGB:**

  RGB is the **primary spatial signal** feeding the model; it's what lets the network "see" the image the way a camera captures it, rather than in a transformed domain. Every video first goes through frame extraction, and each extracted frame, in its raw RGB form, is resized, normalized, and augmented before being passed into the ResNet18 backbone. Since the backbone is frozen everywhere except layer4 and fc, the RGB derived features from pretrained convolutional filters (edges, textures, color patterns) are reused almost entirely only the deepest layers are re trained to specialize in spotting the visual irregularities of AI generated frames, such as unnaturally smooth textures or inconsistent color statistics. Each frame gets its own AI/real prediction, and the video's final label is decided by majority vote across all sampled frames. So, RGB isn't just a single frame decision, it drives the entire chain from raw pixels to the final classification.

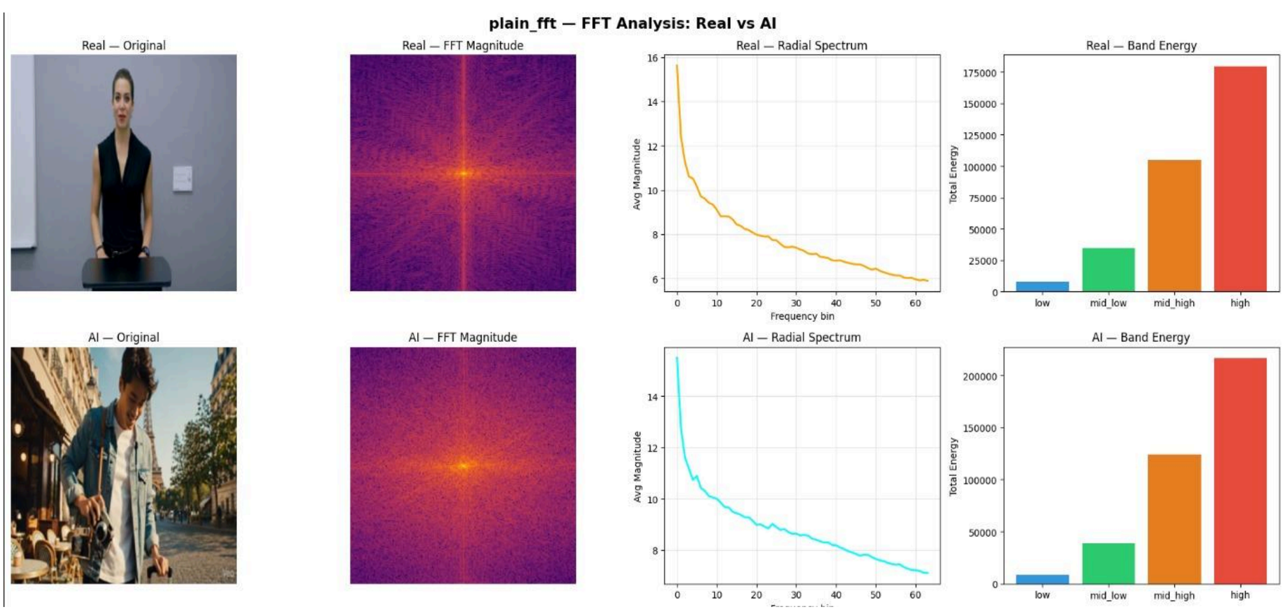
- **Plain FFT:**

  Converts each frame entirely into a 3 channel frequency domain representation

  grayscale FFT magnitude

  green channel FFT magnitude

  grayscale FFT phase



before training, no raw pixel values are retained. then these converted frames further passed on to the ResNet model. The model learns purely from spectral patterns, making it sensitive to frequency domain generator fingerprints but blind to purely spatial artifacts.

- **FFT trained ResNet on RGB:**

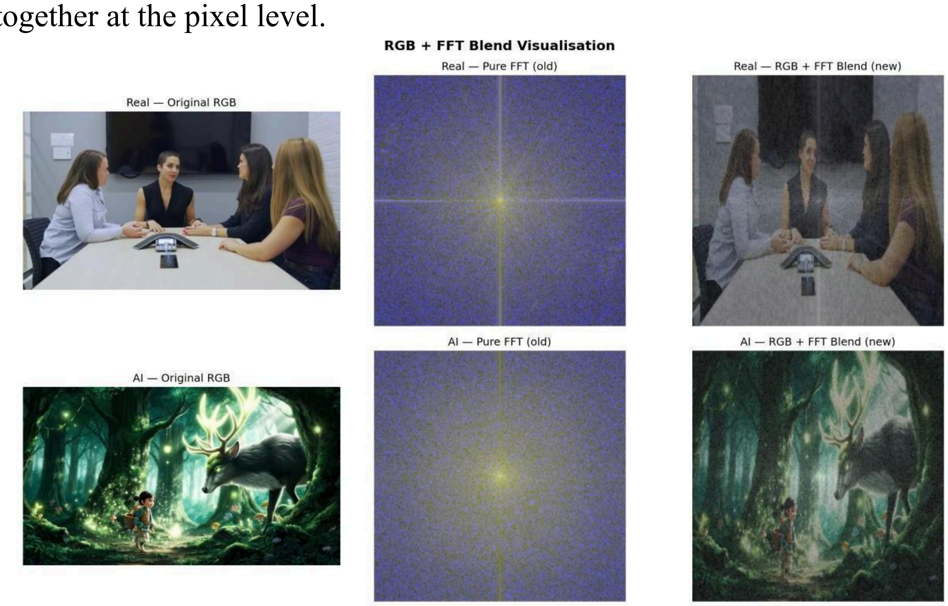
  The FFT trained model was fine tuned specifically on the statistical characteristics of frequency domain data, log scaled magnitude values, a radially symmetric structure dominated by low frequency energy near the center, and spectral peak patterns rather than spatial ones. RGB images have none of these properties: they consist of spatially localized textures and color gradients, with pixel intensities normalized roughly to [0, 1] and a layout that reflects spatial content rather than frequency bins. As a result, feeding RGB frames into the FFT trained model presents it with an input distribution it was never trained to interpret. The convolutional filters which had learned to respond to frequency domain structure, produce activations that carry little to no meaningful signal when applied to RGB inputs.

- **RGB trained ResNet on FFT:**

  The RGB model's layer4 and fc layers were fine tuned specifically on the statistical distribution of RGB pixel values, natural color gradients, texture patterns, pixel intensity ranges roughly in [0,1] after normalization. FFT magnitude spectrograms look nothing like that: they're typically dominated by a bright low frequency center, log scaled energy values, and a completely different spatial layout (frequency bins instead of spatial pixels). So when you feed FFT frames into an RGB trained model, you're essentially asking it to interpret an input distribution it has never seen. The learned filters in layer4 are tuned to detect RGB domain patterns, and those same filters produce close to meaningless activations on frequency domain input.

- **RGB + FFT Blend (α=0.5):**

  Blends each RGB pixel with the frame's normalized FFT magnitude at that location, in equal **50/50 proportion**, producing a single fused 3 channel image. This lets the model learn from both spatial texture and frequency structure simultaneously, at the cost of some information loss from merging the two signals together at the pixel level.



- **Fusion Model (Dual Branch Feature Fusion):**

  Instead of just blending the images together like the **50/50 method** which can actually wash out important details, our dedicated Fusion model uses a "late fusion" (or feature level) approach. We built a dual branch architecture using two separate ResNet 18 backbones running in parallel. One branch handles the raw RGB spatial frames, while the second branch simultaneously processes the FFT frequency frames.

  To make this work, we removed the standard fully connected classification layers from both ResNets so they act purely as feature extractors. The RGB branch looks for visual mistakes like overly smooth textures or color inconsistencies, while the FFT branch hunts for hidden spectral fingerprints like upsampling grids.

  Each branch outputs its own 512 dimensional feature vector. We then concatenate these together into one large 1024 dimensional vector and pass it into a custom Multi Layer Perceptron (MLP) classifier. By combining the data at this deeper level, the model actually learns the complex relationship between what a frame looks like visually and how its underlying **frequency structure** behaves. Finally, just like the plain methods, we take a majority vote across all the sampled frames to make our final "Real" or "AI" prediction for the video.

Overall Workflow

Video Input → Frame Extraction → Preprocessing & Augmentation → Feature Preparation (RGB / FFT / Fusion) → Dataset Creation → ResNet18 Training → Classification → Comparative Evaluation

The same pipeline is followed for all three methods, with differences mainly in the type of features provided to the model.

Step 1: Frame Extraction

Raw videos are processed using OpenCV to extract frames at uniform intervals. Instead of relying on a single frame, multiple representative frames are taken from each video to capture sufficient information for classification. All frames are resized to a fixed resolution to maintain consistency and reduce computational complexity.

These extracted frames serve as input for all three approaches:

- **RGB Method:** Uses original image frames directly.
- **FFT Method:** Uses frequency domain representations generated from the frames.
- **Fusion Method:** Uses both RGB frames and their corresponding FFT representations.

In this project, we perform a comparative study of Spatial, Frequency, and Hybrid Feature Learning for AI Generated Video Detection. Since our objective is to classify whether a given video is AI generated or real using Convolutional Neural Networks (CNNs), processing an entire video directly would be computationally expensive and inefficient. Videos contain hundreds or even thousands of frames, making feature extraction and model training time consuming while also requiring significant computational resources.

To address this, each video is divided into a fixed number of representative frames. These frames are then used as independent training samples for the CNN models. CNNs have demonstrated excellent performance in learning spatial patterns from images; therefore, converting videos into a sequence of images allows the model to learn discriminative features while significantly reducing computational complexity.

Although AI generated videos often contain temporal inconsistencies that image based models may not fully capture, this study investigates how effectively image based CNNs can detect AI generated content using only visual information from individual frames. **Each extracted frame is independently classified, and the predictions are later combined using majority voting to obtain the final prediction for the complete video.** Both frame level and video level performance are evaluated to analyze the effectiveness of the proposed approaches.

To determine an appropriate number of frames for each video, experiments were conducted using 10, 20, 30, and 40 frames per video. Using only 10 frames yielded insufficient visual information, leading to important AI generated artifacts being missed. Increasing the number to 20 frames improved performance, but several videos still lacked adequate scene coverage. Extracting 40 frames provided more information but also introduced additional redundant frames, increasing computational cost without producing a significant improvement in detection accuracy. Based on these observations, 25 frames per video provided the best trade off between computational efficiency and detection performance. It offered sufficient coverage of the video's visual content while avoiding unnecessary redundancy, making it the optimal frame extraction frequency for all subsequent experiments.

Representation of Different Feature Learning Approaches:
RGB (Spatial) Model: Learns spatial features such as textures, colors, edges, facial structures, lighting, and other visual patterns directly from RGB images.
FFT (Frequency) Model: Converts each frame into its frequency domain representation using the Fast Fourier Transform (FFT), allowing the model to learn subtle frequency artifacts and inconsistencies that may not be visible in the spatial domain.
Hybrid Model: Combines both RGB and FFT representations, enabling the model to learn complementary spatial and frequency domain features simultaneously, resulting in a more comprehensive understanding of the visual content.

Step 2: Preprocessing and Augmentation

The extracted frames undergo preprocessing, such as normalization and resizing. For FFT based analysis, frames may also be converted appropriately before frequency transformation.
To improve generalization and increase dataset diversity, augmentation techniques such as:

- Horizontal flipping
- Small rotation
- Brightness adjustments are applied. These steps improve robustness across

varying video conditions

The dataset used in this study consists of 200 videos, with 100 AI generated videos and 100 real videos. After extracting frames, the training dataset contains only a few thousand images, which is relatively small for effectively training a deep Convolutional Neural Network (CNN). Training a CNN on such limited data can lead to overfitting, where the model memorizes the training images instead of learning generalized features that can accurately classify unseen videos.

To overcome this limitation, data augmentation is applied to the training frames. Data augmentation artificially increases the diversity of the training data by generating modified versions of the original images. These modifications include operations such as random resizing, horizontal flipping, slight rotations, color adjustments, blurring, and random erasing. Although these transformations alter the appearance of the images, the underlying class label (AI generated or Real) remains unchanged. By exposing the model to multiple variations of the same frame during training, the CNN learns to focus on meaningful visual patterns rather than memorizing specific images. This makes the model more robust to changes in viewpoint, lighting, scale, orientation, and image quality, ultimately improving its ability to generalize to previously unseen videos.

It is important to note that data augmentation is applied only to the training dataset. The validation and test datasets are not augmented; they are only resized and normalized before being passed to the model. This ensures that the model is evaluated on original, unseen data, providing a fair and unbiased measure of its real world performance. Augmenting only the training data allows the model to benefit from increased data diversity during learning while maintaining the integrity of the evaluation process.

Step 3: Feature Preparation for Three Methods

A central part of this work is the comparative study of three feature representations:

1. RGB Based Method

The RGB stream feeds the model raw spatial pixel information color, texture, and local intensity patterns directly from the frame, without any frequency domain transformation. It's the baseline visual signal that captures how the image looks, as opposed to how its underlying frequency structure behaves.

Working: The classifier's backbone or layers, learns discriminative features from raw pixel level patterns such as color distribution, edge sharpness, texture smoothness, and local pixel correlations. Real frames carry natural sensor noise, demosaicing artifacts, and organic color transitions inherited from camera capture. AI generated frames often show subtly different statistics here, including over smoothed textures, unnaturally consistent color, or missing sensor level noise, which the convolutional layers learn to pick up on during training.

RGB based detection is generator specific, since artifacts vary considerably across generation methods like GANs versus diffusion models versus newer architectures, so a classifier trained on one generator's visual signature may not generalize well to another.

It's also sensitive to compression, because video codec re-encoding tends to smooth textures and alter color and noise statistics, washing out the fine grained pixel level cues RGB depends on. Used in isolation, RGB also misses periodic or frequency based fingerprints entirely. Finally, RGB texture cues can shift with frame resizing or resolution changes, so consistent preprocessing between train and test sets is needed to avoid distribution mismatch.

2. FFT Based Method

The FFT stream feeds the model a frequency domain representation of the frame; rather than raw spatial pixels, it converts the image via the 2D Fourier Transform

$$F(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y)\, e^{-i2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}$$

then computes log compressed magnitude of that spectrum as the actual input channels. Instead of "what does this frame look like," the model is now looking at "how much of each spatial frequency & how fast intensity changes across the image is present, and how strongly."

- Low frequencies capture smooth, slowly varying regions;
- High frequencies capture edges, fine texture, noise, and repeating fine grained patterns

Working: The classifier's backbone learns discriminative features from the shape and structure of this frequency spectrum rather than pixel colors or edges directly. Real camera captured frames tend to show a roughly natural, smoothly decaying frequency profile, with energy concentrated in low to mid frequencies, with an organic falloff toward high frequencies driven by sensor noise and lens characteristics. AI generated frames, by contrast, often carry periodic, generator specific fingerprints baked in by the upsampling or synthesis process itself, GAN checkerboard artifacts from transposed convolutions, or diffusion model's characteristic high frequency grid patterns which show up as distinct peaks, grid like structures, or unnatural energy concentrations in the spectrum that convolutional layers can learn to key on, even when the equivalent RGB frame looks visually convincing.
This is precisely why FFT is a useful

3. Fusion (RGB + FFT) Method

The Fusion stream utilizes a dual branch architecture that simultaneously ingests both the raw spatial pixels (RGB) and the frequency domain spectrum (FFT). Instead of forcing the model to rely solely on "what the frame looks like" or "how its underlying frequencies behave," this method feeds both representations into parallel networks, effectively creating a unified, multi-modal view of the data.

Working: The classifier operates using two independent backbones (e.g., parallel ResNet networks) that extract features simultaneously. The RGB branch learns to identify spatial anomalies such as unnatural lighting, blending errors, blurry boundaries, or anatomical glitches. Concurrently, the FFT branch learns to detect hidden periodic artifacts such as upsampling grids, GAN checkerboard patterns, or unnatural high frequency energy peaks that are entirely invisible to the naked eye.
Mathematically, the network extracts a high dimensional feature vector from each domain (for example, a 512 dimensional vector from RGB and a 512 dimensional vector from FFT). At the "fusion layer," these two vectors are concatenated into a single, massive feature space (a 1024 dimensional combined vector). This concatenated tensor is then passed through a Multi Layer Perceptron (MLP) classifier head. This allows the model to learn the nonlinear cross correlations between the two domains, essentially figuring out how a specific visual glitch in the spatial domain correlates with a specific frequency spike in the spectral domain to make a final "Real" or "AI" prediction.

Step 4: Dataset Creation

A custom PyTorch Dataset is used to manage the three input settings. Depending on the experiment, it loads:

- RGB frames only
- FFT representations only
- Both RGB and FFT inputs for fusion

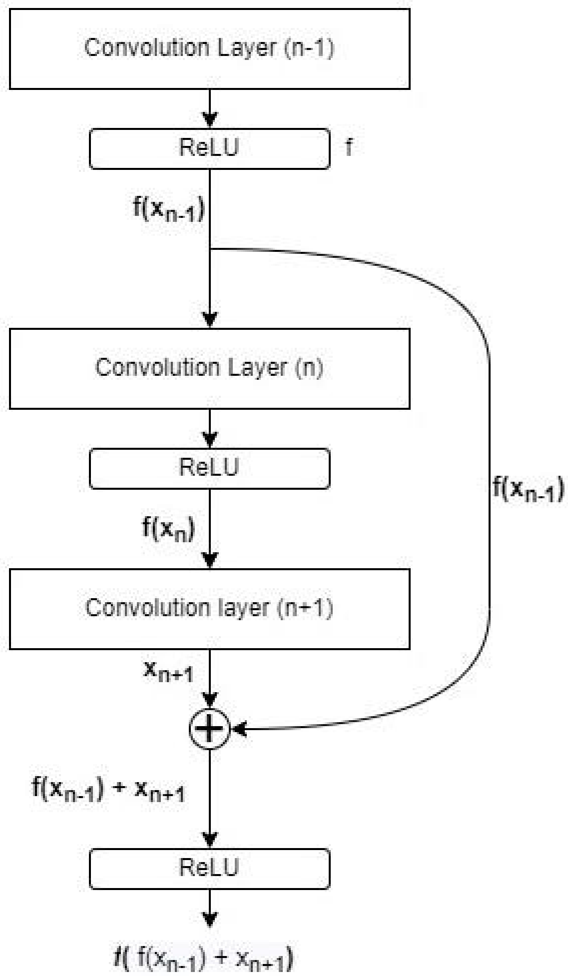
To build this dataset, we collected a total of 200 videos, evenly split between 100 AI generated and 100 real videos. These videos were deliberately sourced from a wide variety of origins rather than a single generator or camera source, so that the dataset reflects a realistic range of quality levels rather than a narrow, idealized slice of data. This diversity was an intentional design choice: a model trained only on clean, high quality samples tends to perform well in controlled settings but struggles when faced with the messier, more varied content it would actually encounter in the real world.

Within the AI generated portion, we also made sure to include a small number of videos produced by the most advanced, high end generative models currently available. These represent the hardest cases for detection, since state of the art generators leave far fewer visible or spectral artifacts than older or lower quality tools. Including even a handful of these samples pushes the model to learn subtler, more robust cues rather than relying on obvious low quality calls.

The 1/0 labeling convention (1 for AI, 0 for real) is used primarily to support clear interpretation during the testing and evaluation phase, making it straightforward to read confusion matrices, classification reports, and per video prediction outputs. Every video in the dataset is stored under a unique filename, with no duplication across samples. This was done to ensure the model is always learning from genuinely distinct examples, preventing it from memorizing repeated inputs and giving a more honest measure of how well it generalizes. The dataset is split into training, validation, and testing sets for systematic evaluation.

Step 5: Model Architecture

The classification model is built using **ResNet18** with pretrained ImageNet weights. Its final layer is modified for binary classification.



The model is used across all three methods:

- For **RGB**, ResNet18 learns spatial features.
- For **FFT**, ResNet18 learns spectral patterns from transformed frequency inputs.
- For **Fusion**, RGB and FFT features are combined through joint processing or feature concatenation before classification.

ResNet was chosen as the backbone because its residual connections allow deep networks to train effectively without suffering from vanishing gradients, which makes it well suited for learning the subtle, layered patterns needed to distinguish AI generated frames from real ones. It also comes with strong pretrained weights from large scale image datasets, giving the model a solid foundation of general visual features (edges, textures, shapes) that can be fine tuned rather than learned from scratch, which is especially valuable given our relatively small dataset of 200 videos. Additionally, ResNet's architecture is lightweight enough to train efficiently on Colab while still being deep enough to capture the fine grained spatial and spectral artifacts that separate synthetic content from real footage.

Step 6: Training Strategy

A common training strategy is adopted across all three approaches to ensure consistency and fair comparison. The models are trained using standard deep learning optimization techniques aimed at improving convergence, generalization, and stability.

The training process includes an appropriate loss function for binary classification, an optimizer for effective weight updates, learning rate scheduling to support smoother convergence, and regularization strategies to reduce overfitting. Additional stabilization techniques are incorporated where necessary to improve training reliability.

Applying a unified training strategy across the **RGB**, **FFT**, and **Fusion** models ensures that performance differences arise from the feature representations themselves rather than differences in training methodology.

Step 7: Evaluation and Comparative Analysis

The trained models are evaluated using performance metrics such as:

- Accuracy
- Loss trends
- Confusion Matrix
- Comparative performance analysis

A direct comparison is then performed between:

1. **RGB only model** – spatial feature learning
2. **FFT only model** – frequency domain feature learning
3. **FFT+RGB mixed feature model –** extracting (0.5*features) from both fft&rgb to train a single ResNet
4. **Fusion model (RGB + FFT)** – combined spatial and spectral learning

This comparative study forms the core of the methodology, helping analyze the strengths and limitations of each method. While RGB captures visible visual cues and FFT captures hidden spectral artifacts, the fusion approach is expected to provide the most robust performance by leveraging both representations together.

Thus, the proposed workflow not only develops an AI generated video detection system but also systematically evaluates how the three chosen methods contribute to improving detection accuracy and reliability.

7. Libraries and Implementation

The implementation of this project was carried out using Python, supported by a range of libraries for deep learning, image processing, data handling, visualization, and model evaluation. These libraries collectively enabled the development and comparison of the three detection approaches used in this study: **RGB based detection, FFT based detection, and Fusion (RGB + FFT) detection (version 1 2).**

Core Programming and Utility Libraries

Standard Python libraries such as **os**, **pathlib**, **copy**, **io**, **re**, and **random** were used for file handling, dataset organization, path management, random sampling, and managing model checkpoints during training. The **collections** module, particularly **Counter** and **defaultdict**, was used for examining class distribution and organizing grouped data within the dataset. **warnings** were used to suppress non critical runtime warnings for cleaner output, and **tqdm** provided progress bars during training and evaluation loops, making it easier to track long running processes. NumPy was used extensively for numerical operations, array manipulation, and supporting the FFT computations required for frequency domain analysis.

Deep Learning Libraries

The core implementation was built using PyTorch, which served as the primary deep learning framework for model development and training. Modules from **torch.nn** were used to define and modify the neural network architecture, while **torch.optim** supported optimization during training, with **CosineAnnealingLR** from **torch.optim.lr_scheduler** used to schedule the learning rate across epochs for smoother convergence.

Torchvision played a major role in this project, particularly through its **models** module, which provided access to the pretrained ResNet18 architecture used across all three methods, and its **transforms** module, which handled image preprocessing and augmentation for RGB, FFT, and fusion based inputs alike. **DataLoader** and custom **Dataset** classes from **torch.utils.data** were used to efficiently manage batching and loading of training, validation, and test data across all three experimental settings.

Image and Video Processing Libraries

For video processing and frame extraction, OpenCV (**cv2**) was used to read videos and extract frames at regular intervals, which served as the raw input for all three approaches. PIL (**Image**) was used for image loading and preprocessing before data was passed into the deep learning pipeline. For the FFT based method, NumPy based frequency transformations were used to generate frequency domain representations from the extracted frames, which were also combined with RGB inputs in the fusion approach to enable direct comparison across all three methods.

Evaluation Libraries

Scikit learn was used for model evaluation and dataset splitting. **train_test_split** handled the division of data into training, validation, and test sets, while **accuracy_score**, **precision_score**, **recall_score**, **f1_score**, **classification_report**, and **confusion_matrix** were used to assess and compare the performance of the RGB, FFT, and Fusion models in detail.

Visualization Libraries

For analyzing results and presenting findings, Matplotlib (including **pyplot** and **patches**) and Seaborn were used to plot training and validation curves, accuracy comparisons, confusion matrices, and comparative performance visualizations across the three methods. These visualizations were essential in interpreting the relative strengths and limitations of each approach.

Implementation Overview

The implementation was designed around a unified pipeline where the same deep learning framework and model backbone (**ResNet18**) were used for all three approaches to ensure fair comparison.

RGB method:

In the **RGB method**, the model was trained using original video frames.

The ResNet18 build was done by freezing everything and unfreezing the last residual block.

```
def build_resnet18():
    model = models.resnet18(
        weights=models.ResNet18_Weights.IMAGENET1K_V1
    )
    # Freeze everything
    for param in model.parameters():
        param.requires_grad = False
    # Unfreeze last residual block
    for param in model.layer4.parameters():
        param.requires_grad = True
    model.fc = nn.Linear(model.fc.in_features, 2)
    trainable = sum(p.numel() for p in model.parameters() if
p.requires_grad)
    total = sum(p.numel() for p in model.parameters())
    print(f"Trainable params: {trainable:,} / {total:,}")
    return model.to(DEVICE)
```

To improve generalization and reduce overfitting, the following augmentation pipeline is applied to the RGB frames during training.

```
def get_train_transforms():
    return transforms.Compose([
        transforms.Resize((IMG_SIZE, IMG_SIZE)),
        transforms.RandomHorizontalFlip(p=0.5),
        transforms.RandomRotation(degrees=10),
        transforms.ColorJitter(brightness=0.3, contrast=0.3,
saturation=0.2),
        transforms.RandomResizedCrop(IMG_SIZE, scale=(0.8, 1.0)),
       transforms.RandomApply([transforms.GaussianBlur(kernel_size=3)],
p=0.2),
        transforms.ToTensor(),
        transforms.RandomErasing(p=0.2, scale=(0.02, 0.1)),
        transforms.Normalize(mean=[0.485, 0.456, 0.406],
                             std=[0.229, 0.224, 0.225]),
    ])
def get_val_transforms():
    return transforms.Compose([
        transforms.Resize((IMG_SIZE, IMG_SIZE)),
        transforms.ToTensor(),
        transforms.Normalize(mean=[0.485, 0.456, 0.406],
                             std=[0.229, 0.224, 0.225]),
    ])
```

FFT method:

Converting regular pixels into frequency domain and normalising to the range between 0 and 1 for computational ease

```
def fft_to_3channel(img_rgb_np,img_size=IMG_SIZE,
log_scale=FFT_LOG_SCALE):
    img = img_rgb_np.astype(np.float32)
    gray = 0.299 * img[:, :, 0] + 0.587 * img[:, :, 1] + 0.114 * img[:,
:, 2]
    fft_g = np.fft.fftshift(np.fft.fft2(gray))
    fft_G = np.fft.fftshift(np.fft.fft2(img[:, :, 1]))
    mag_g = np.log1p(np.abs(fft_g)) if log_scale else np.abs(fft_g)
    mag_G = np.log1p(np.abs(fft_G)) if log_scale else np.abs(fft_G)
    phase = (np.angle(fft_g) + np.pi) / (2 * np.pi)

    return np.stack([_nr(mag_g, img_size), _nr(mag_G, img_size),
_nr(phase, img_size)], axis=0)
```

FFT+RGB 0.5 frames method:

Extracting both the features simultaneously instead of plain FFT or RGB and normalizing it 0 1 scale for computational ease and passing the results for further analysis using the ResNet.

```
def rgb_fft_blend(img_rgb_np, img_size=IMG_SIZE, alpha=RGB_ALPHA,
log_scale=FFT_LOG_SCALE):
    img = img_rgb_np.astype(np.float32) / 255.0
    img = cv2.resize(img, (img_size, img_size))
    gray = 0.299 * img[:, :, 0] + 0.587 * img[:, :, 1] + 0.114 * img[:,
:, 2]
    fft = np.fft.fftshift(np.fft.fft2(gray))
    mag = np.log1p(np.abs(fft)) if log_scale else np.abs(fft)
    mag = (mag   mag.min()) / (mag.max() + 1e 8)
    mag = cv2.resize(mag, (img_size, img_size))
    blended = img.copy()
    for c in range(3):
        blended[:, :, c] = (1   alpha) * img[:, :, c] + alpha * mag

    return np.clip(blended, 0, 1).transpose(2, 0, 1).astype(np.float32)
```

Fusion model (RGB + FFT):

Dual Feature Extractor Architecture (Parallel CNNs):

```
class HybridResNet(nn.Module):
    def __init__(self, num_classes=2):
        super(HybridResNet, self).__init__()

       self.rgb_branch=models.resnet18(weights=models.ResNet18_Weights.
DEFAULT)
        num_ftrs_rgb = self.rgb_branch.fc.in_features
        self.rgb_branch.fc = nn.Identity()

        self.fft_branch =
models.resnet18(weights=models.ResNet18_Weights.DEFAULT)
        num_ftrs_fft = self.fft_branch.fc.in_features
        self.fft_branch.fc = nn.Identity()
```

The Fusion Classifier Head:

It takes the 512 dimensional outputs from the RGB and FFT branches, concatenates them into a single 1024 dimensional tensor, and passes it through a Multi Layer Perceptron (MLP) to make the final Real vs. AI decision.

```
        # Fusion Classifier Head (MLP)
        self.classifier = nn.Sequential(
            nn.Linear(num_ftrs_rgb + num_ftrs_fft, 512),
            nn.ReLU(),
            nn.Dropout(0.5),
            nn.Linear(512, num_classes)
        )
 def forward(self, rgb, fft):
       # Extract features from both domains simultaneously
       feat_rgb = self.rgb_branch(rgb)
       feat_fft = self.fft_branch(fft)
       # Concatenate and classify
       return self.classifier(torch.cat((feat_rgb, feat_fft),
dim=1))
```

Spatial Transformation Block (RGB):

```
# Standard ImageNet Transforms for the RGB Branch
transform = transforms.Compose([
    transforms.Resize((224, 224)),
    transforms.ToTensor(),
    transforms.Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224,
0.225])
])
# Applying the transform to an extracted frame
rgb_img = Image.fromarray(cv2.cvtColor(frame, cv2.COLOR_BGR2RGB))
             frames_rgb.append(transform(rgb_img))
```

Frequency Transformation Block (FFT):

It converts the spatial frame to grayscale, applies the 2D Fast Fourier Transform, shifts the low frequencies to the center, and applies logarithmic compression to reveal the hidden AI artifacts.

```
 gray    = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
            f_shift = np.fft.fftshift(np.fft.fft2(gray))
            mag     = 20 * np.log(np.abs(f_shift) + 1e 8)
            mag     = cv2.normalize(mag, None, 0, 255, cv2.NORM_MINMAX)
            fft_img = Image.fromarray(np.uint8(mag)).convert('RGB')
            frames_fft.append(transform(fft_img)
```

Loss Function and Optimizer:

CrossEntropyLoss is used as the loss function, suited for this binary classification task. The Adam optimizer is applied only to the trainable parameters (layer4 and fc), with weight decay added for regularization to reduce overfitting. A StepLR scheduler decays the learning rate by a factor of 0.1 every 7 epochs, allowing larger updates early in training and finer adjustments as the model converge

8. Results and Findings

FrameLevel Metrics			
Model	Accuracy	Macro Precision	Macro Recall
Plain RGB	83%	84%	83%
RGB+FFT Blend (α=0.5)	88%	88%	88%
RGB+FFT Fusion	88.79%	89.28%	88.78%

VideoLevel Metrics			
Model	Accuracy	Macro Precision	Macro Recall
Plain RGB	87%	87%	86.5%
RGB+FFT Blend (α=0.5)	90%	90.18%	90%
RGB+FFT Fusion	90%	90.18%	90%

Findings:

Some key findings from the comparative study include:

- RGB based detection effectively captures visible spatial artifacts but may miss hidden generative traces.
- FFT based detection reveals spectral patterns associated with synthetic generation and improves detection beyond spatial analysis alone.
- Fusion based detection performs better than individual methods by leveraging both feature domains.
- Frequency domain information plays a significant role in improving robustness, especially for challenging synthetic samples.

9. Comparative Study

Across the three approaches, model performance varied markedly. The Plain RGB model achieved 87% accuracy, the RGB+FFT Blend model achieved 90%, while the RGB+FFT Fusion model performed the best with 90% accuracy. These differences can be explained by how each approach utilizes spatial and frequency domain information.

The Plain RGB model relies entirely on spatial information present in the image, such as color, texture, and intensity patterns. Since it only analyzes the pixel values, it learns to identify visual artifacts commonly introduced by AI generated content, including unnatural smoothness, inconsistent textures, or subtle color irregularities. However, because it has no access to frequency domain information, it may miss patterns that are not easily visible in the spatial domain.

The RGB+FFT Blend approach combines both spatial (RGB) and frequency domain (FFT) information using a fixed weighting factor of $\alpha = 0.5$, allowing the model to leverage complementary features from both representations. By incorporating frequency domain cues alongside spatial features, the model gains access to additional information that is not available in the Plain RGB approach, improving its ability to detect subtle artifacts present in synthetic images. This demonstrates that integrating spectral information is beneficial compared to relying solely on spatial features. However, because the contribution of RGB and FFT features remains fixed for every input image, the model cannot adaptively emphasize the more informative representation when image characteristics vary. Consequently, although the RGB+FFT Blend approach performs better than the single feature Plain RGB model by exploiting complementary information, its lack of adaptive feature weighting limits its effectiveness compared to the RGB+FFT Fusion approach, which learns to combine spatial and spectral features dynamically.

In contrast, the RGB+FFT Fusion model allows the network to learn how to combine spatial and spectral information during training. Instead of assigning equal importance to RGB and FFT representations, the model automatically determines which features are more useful for each input. This adaptive fusion enables it to emphasize the most discriminative information while suppressing less relevant features, allowing it to capture complementary patterns from both domains more effectively.

Overall, the key difference between the three approaches lies in how they utilize spatial and frequency domain information. The Plain RGB model relies solely on spatial features, whereas both the RGB+FFT Blend and RGB+FFT Fusion models exploit complementary information from the RGB and FFT domains, enabling them to capture a broader range of discriminative features. The RGB+FFT Blend approach demonstrates that integrating spatial and spectral representations improves the model's ability to detect synthetic content compared to using spatial information alone. Building on this, the RGB+FFT Fusion model further enhances performance by learning a more effective interaction between the two feature representations, achieving the highest accuracy of 90%. These results indicate that incorporating both spatial and frequency domain features is beneficial, while a learnable fusion mechanism provides the greatest improvement in classification performance.

Ensemble Strategy: Majority Voting

To improve robustness beyond any single architecture, the final system combines predictions from all three models using majority voting, where the class receiving the most individual model votes is taken as the final classification.The theoretical basis for this approach follows from treating each model as an independent classifier with accuracy P_i. Under the assumption of independent errors, the ensemble is incorrect only when a majority (at least 2 of 3) of the models are simultaneously wrong on the same input, a substantially rarer event than any single model failing. Formally, if X_i is a Bernoulli random variable indicating whether model i classifies correctly:

$$\mathbf{P(ensemble\ correct)=P(\textstyle\sum^{3}_{i=0}X_i\geq 2)}$$

Limitations:

1. Limited Dataset:

The project was trained and evaluated on only 200 videos (100 AI generated and 100 real), which is a relatively small dataset for a deep learning based detection task. While efforts were made to include diverse sources and quality levels, this limited sample size restricts how well the model can learn robust, generalizable patterns, and increases the risk of the model picking up on dataset specific quirks rather than truly universal indicators of synthetic content.

2. Rapidly Advancing GAN and Generative Models:

The generative AI landscape is evolving extremely fast, with new models consistently producing more realistic and artifact free content than the ones used to build this dataset. High end generative models available in the market today are often far more advanced than what could be fully represented in our training data, meaning the model may struggle to detect content from cutting edge generators that leave minimal or no visible spatial and frequency domain traces. This is an ongoing challenge for any detection system, since the underlying problem is a constantly moving target.

3. Computational Power:

Training and experimentation were constrained by the computational resources available through Google Colab. This limited the scale of experiments possible, including batch size, the number of models that could be trained in parallel, and how extensively hyperparameters could be tuned. Working with a frozen backbone and fine tuning only layer4 and fc was partly a necessity shaped by these compute constraints, rather than purely a design choice.

4. Learning Rate and Epoch Constraints:

Due to time and resource limitations, training was capped at 10 epochs with early stopping applied at a patience of 3. While this helped prevent excessive overfitting and kept training times manageable, it also means the model may not have fully converged to its best possible performance. A longer training schedule with more careful learning rate tuning could potentially yield better results, but was not feasible within the scope and resources of this project.

5. Overfitting:
Despite freezing most of the backbone and applying augmentation, signs of overfitting were still observed during training. This is likely a direct consequence of the limited dataset size combined with the relatively small number of trainable parameters being fine tuned on a narrow set of examples. The gap between training and validation performance in some runs indicated that the model was, to some extent, memorizing patterns specific to the training data rather than learning fully generalizable detection cues.

Scope for Improvement

While the proposed system demonstrates promising results, there remains significant room for improvement.

One of the key limitations of this work is that the current framework relies primarily on frame level classification using ResNet18, which is effective at capturing strong spatial and spectral features but has limited ability to model temporal inconsistencies across a video sequence. Since videos inherently carry motion based patterns that can reveal signs of synthetic manipulation, incorporating temporal information into the pipeline could meaningfully improve detection performance going forward.

To push detection accuracy further, future work could integrate more advanced temporal architectures:

- **LSTM (Long Short Term Memory) networks:** or modeling how frames evolve over time, capturing sequential dependencies
- **RNN based models:** for capturing frame to frame dependencies that a purely frame independent approach misses
- **3D CNN architectures:** for learning spatial and temporal features jointly rather than separately.
- **Transformer based or hybrid models:** for building richer, more context aware video representations

Together, these approaches could help the model pick up on cues that frame based methods tend to overlook motion irregularities, unnatural transitions between frames, blinking inconsistencies, and other subtle temporal artifacts that only become visible when a video is understood as a sequence rather than a set of isolated frames.
Beyond architectural upgrades, several other directions could strengthen the system further:

- **Larger and more diverse training datasets:** to help the model generalize better across different generative sources and video qualities
- **More advanced fusion strategies:** to make better use of RGB and FFT information together

- **Stronger backbone architectures beyond ResNet18:** to improve the model's representational capacity
- **Real time optimization and deployment oriented models:** to make the system more practical for actual real world use rather than offline evaluation alone

In its current form, the project produces encouraging, moderate to moderately high results. Just as importantly, it lays a solid foundation for the improvements outlined above. Incorporating temporal deep learning models such as LSTM and related architectures holds strong potential to push this work toward significantly higher accuracy in future iterations.

A comparative study was conducted using the three selected approaches **RGB based detection, FFT based detection, and Fusion (RGB + FFT) detection** using the same dataset split and training pipeline to ensure fair evaluation.

The **RGB only model** provided moderate performance and was effective in detecting visible spatial artifacts, but its performance reduced when synthetic videos appeared highly realistic.

The **FFT only model** showed improved discrimination by capturing hidden frequency domain artifacts introduced during content generation. However, relying only on spectral information may miss some spatial details.
Among the three methods, the **Fusion (RGB + FFT) model** produced the best overall performance by combining spatial and frequency domain features. This approach offered better robustness and accuracy compared to using either RGB or FFT alone, though with slightly higher computational complexity.

Overall comparison shows:

- **RGB only:** Moderate baseline performance using spatial features
- **FFT only:** Better detection through spectral analysis
- **Fusion (RGB+FFT):** Best performance through combined feature learning

The results suggest that the fusion approach provides the best trade off between accuracy and robustness. However, the achieved performance is moderate to moderately high, and further improvements are possible using temporal models such as **LSTM, RNN, or 3DCNNs**, which could help achieve even higher detection accuracy.

10. Conclusion

This project demonstrates that combining frequency domain analysis with deep learning can improve the detection of AI generated videos compared to relying solely on spatial information. Through a comparative study involving RGB based detection, FFT based detection, and Fusion (RGB + FFT) detection, the work shows how different feature representations contribute to distinguishing real and synthetic media.

The RGB only method provided a solid baseline through spatial feature learning, while the FFT based method showed the value of frequency domain analysis in capturing hidden artifacts introduced by generative models. Among the three approaches, the

Fusion model produced the best overall performance by combining complementary spatial and spectral information, making it the most effective approach in this study.

The results indicate that frequency domain features play an important role in strengthening AI generated media detection, and integrating them with conventional RGB analysis leads to improved robustness and classification reliability.

At the same time, this work also shows that the current framework produces moderate to moderately high performance, while leaving room for further improvements. Since this project primarily uses frame level analysis with ResNet18, incorporating temporal

models such as LSTM, RNN, or 3D CNN architectures could help achieve higher detection accuracy by modeling video level inconsistencies and motion patterns.

Future work can focus on:

- Integrating advanced temporal and hybrid architectures
- Evaluating on larger and more diverse datasets
- Exploring stronger fusion strategies
- Optimizing models for real time or resource constrained deployment

Overall, this project establishes that combining RGB and FFT representations is a promising direction for AI generated video detection and provides a foundation for future improvements toward more accurate and practical deepfake detection systems.

Reference: This project is based on the study done on the research paper;
Beyond Deepfake Images: Detecting AI Generated Videos